

contents

ELECTRONICS FOR YOU PLUS | NOVEMBER 2015 | VOL. 04 NO. 07 | ISSN-2454-4426



TECH FOCUS

56 Renewable Energy: From Sugar-Powered Battery to Clean Tech Inspired by Butterflies

BUYERS' GUIDE

84 How To Select Your Own Benchtop SMD Assembly



20 Healthcare

Bioinformatics: A Shot of Life to Information Technology

26 Power Supply

Harvesting Radio Frequency Energy

32 Cyber Crime

Digital Forensics and Investigation (Part 1 of 2)

36 Embedded

How Microcontrollers are Getting Better and Better

Internet of Things

42 Connected Systems Require Hardware Based Security

46 Demystifying the IoT for Embedded Design Engineers

52 Test & Measurement

Classic Improvements in Power Supplies

60 Tech Focus

Connectivity and Consumerisation Drive Renewables

66 Defence

American Carrier Strike Groups: An Electronic Perspective (Part 3 of 5)

74 Components

Is Disruption On The Cards For Passives

81 Innovation

A One-Stop Shop for the Custom-Made LTE Stack

88 Interview

Internet of Things: "The industrial segment is currently the fastest growing in the IoT segment" — John Dixon, director - corporate marketing, Freescale

DO-IT-YOURSELF



- Electronics as a Hobby 78
- ATmega16A Based GPS Receiver 106
- Android Based Invisible Broken-Wire-Detection Robot 110
- Automatic USB-Controlled Power Switch 114
- Mains Supply Interruption Alarm 116
- Dual Power Supply and Battery Charger 118
- Power Bank for Smartphones 120
- FreeRTOS on PIC18F4520 MCU 122
- Android Application: Parallel RLC Circuit (Part 2 of 2) 126

EFY Plus DVD

- Yosys: Your Solution for Synthesis I
- FieldTrip: An Open Source MATLAB Toolbox for Biomedical Signal Analysis IV
- Design Digital Circuits Logically VII

90 Interview

System Design: "The challenge has moved from designing a board to its fabrication" — Saugat Sen, vice president, research and development, Silicon-Package-Board R&D Group, Cadence

92 EFY Plus DVD

This Month's DVD Contents

98 Make in India

Market Survey: Industrial Automation: Enabling the Next Industrial Revolution

eStyle

132 Buyers' Guide: Buying a High-Priced Smartphone Made Easy

134 DIY: Top Eight Features of Microsoft Office 2016

EDITOR	: Ramesh Chopra
EDITORIAL	: Editorial Secretary
CORRESPONDENCE	: Phone: 011-26810601; E-mail: editsec@efy.in (Technical queries: eflylab@efy.in)
SUBSCRIPTIONS & MISSING ISSUES	: Phone: 011-26810601 or 02 or 03 E-mail: support@efy.in
BACK ISSUES, BOOKS, CDs, PCBs, etc.	: KITS'n'Spares, New Delhi Phone: 011-26371661, 26371662 E-mail: info@kitsnspares.com
EXCLUSIVE NEWSSTAND DISTRIBUTOR	: IBH Books & Magazines Distributors Ltd Unit No. 10, Bezzola Complex, Opp. Saman Nagar, Sanjivnagar Road, Chembur, Mumbai 400 071 Phone: 022-40497401/02
ADVERTISEMENTS	: Phone: 011-26810601 or 02 or 03 E-mail: efymag@efy.in
DELHI (HEAD OFFICE)	: Phone: 022-24950047, 24928520 E-mail: efymum@efy.in
BENGALURU	: Phone: 080-25260394, 25260023 E-mail: efylbr@efy.in
CHENNAI	: Phone: 044-42994363 E-mail: efychen@efy.in
HYDERABAD	: Phone: 040-67172633 E-mail: efymyd@efy.in
PUNE	: Phone: 08800295610, 09870682995 E-mail: efypune@efy.in
GUJARAT	: Phone: 079-61344948 E-mail: efylad@efy.in
CHINA	: Power Pioneer Group Inc. Ph: (86 755) 83729797, (86) 13923802595 E-mail: powerpioneer@efy.in
JAPAN	: Tandem Inc., Ph: 81-3-3541-4166 E-mail: tandem@efy.in
SINGAPORE	: Publicitas Singapore Pte Ltd Ph: +65-6836 2272 E-mail: publicitas@efy.in
TAIWAN	: J.K. Media, Ph: 886-2-87726780 ext. 10 E-mail: jkmedia@efy.in
UNITED STATES	: E & Tech Media Ph: +1 800 536 6677 E-mail: veronique@marque@gmail.com

SUBSCRIPTION RATES

Period	Newstand Price	You Pay	Digital	Overseas	Print
Year	(₹)	(₹)	(₹)	US\$ 15	US\$ 120
Two	2400	1800	750	US\$ 15	—
One	1200	960	280	US\$ 5	US\$ 120

Please send payments only in favour of **EFY Enterprises Pvt Ltd**

Printed, published and owned by Ramesh Chopra. Printed at International Print-o-Pack Ltd, C-404 C-11, Haryana Complex, Phase II Extension, HODDA, 201 305, Gurgaon Road, Gurgaon, Haryana, India. On the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-I, New Delhi 110020. Copyright 2015. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

Regulars

- 06 Feedback
- 08 Q&A
- 10 Useful Websites
- 14 Tech News
- 94 Make in India: Industry News
- 102 New Products
- 130 First Look
- 135 Business Pages Ads
- 149 Electronics Mart Ads
- 154 Product Categories Index + Attractions During 2015
- 155 Advertisers' Index

NEXT ISSUE

- Smart Lighting
- LCD Display Components
- Programmable Power Source